

Dec 01 2025

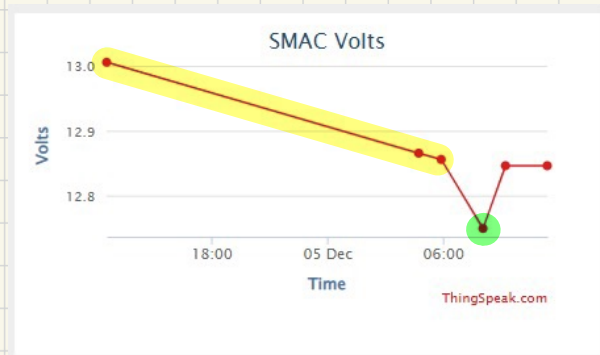
SMAC Cam

Todo this week:

- battery testing with Lipo
- Maybe put RTC with coin battery on PCB for turning off board and buck

Now that I have a battery monitor on the board it should be easier to conduct this test

With the monitor + a Thingspeak dashboard



Early data is weird

I do not know what caused the initial drop here because the device was not turned on. Even weirder is **THIS**! Which lead to a lot of concern, but its... uh... better?

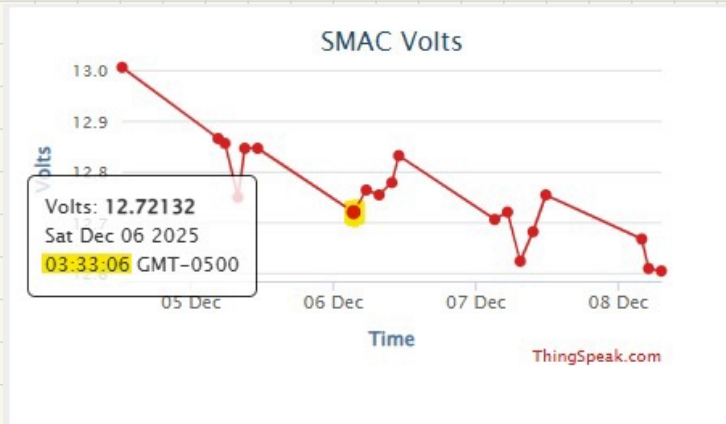
* Note: the solar panel is attached BUT it is inside *

The battery is now outside at a crisp 50-40°F so a voltage drop is expected.

Dec 09 2025

SMAC Cam

I have left SMAC out to continue monitoring how the new battery acts and here is what the data says



★ Note: here the device is not getting any solar so a downward trend is expected.

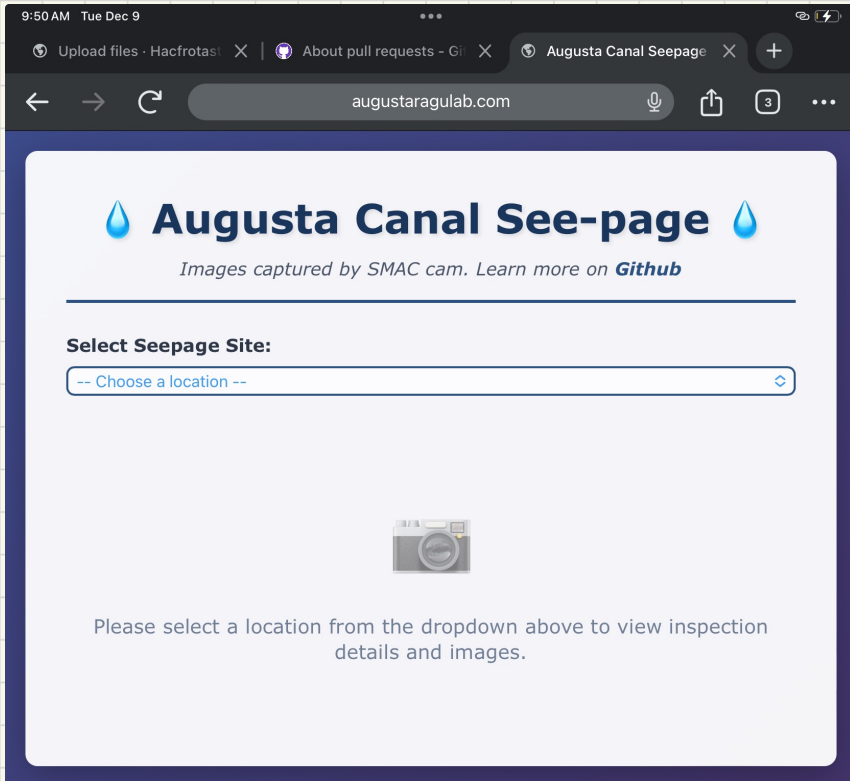
Observations

- SMAC is coming on at the **WRONG TIME!**
Why? ⇒ I forgot to tell the device what time zone it's in
Adding Time.zone(-4) should fix that
- The voltage levels are fluctuating more than I expected
Since there is no solar, why is it going up and down?
⇒ I think this is caused by 2 things
 - 1) due to the time of day (3:00 AM) the IR LEDs should have come on with the device. These have a high current demand and may have lead to a temp. voltage dip
 - 2) It's also winter so the colder weather is also affecting our voltage. Additionally, the device is operating at (what usually is) the coldest part of the day.

Dec 9th 2025

SMAC Cam

I gave the website a bit of a face lift. Now `augustaragulab.com` (or `seepage.augustaragulab.com`) features a drop down for each location. Once a site is selected it displays a brief description and the photo from SMAC



- Its also mobile friendly (which was not the case before)

The html for this page was made by Claude code (Sonnet 4.5)

★ html is very white space UN-friendly --

Dec 9th 2025

SMAC Cam

9:51AM Tue Dec 9

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augustaragulab.com

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🔹 Augusta Canal See-page 🔹

Images captured by SMAC cam. Learn more on [Github](#)

Select Seepage Site:

Aqueduct Park

Aqueduct Park

There are multiple points in the aqueduct that are seeping. We have combined these flows into one and are now able to gather seepage information by looking at one point shown below.

